

Touchless Alcohol Dispenser

with Water Level Indicator Using Arduino

Capabilities

This device provides a completely touchless method of sanitization, helping reduce the spread of germs through physical contact. It automatically controls the amount of alcohol dispensed per use to minimize waste and maintains a steady supply by activating an LED indicator when the liquid level is low. The system is designed for continuous and reliable operation in high-traffic areas such as schools, offices, or home entrances.

Core Components

- Arduino Uno: The central microcontroller board.
- Ultrasonic Sensor (HC-SR04): For non-contact distance measurement.
- Liquid Level Sensor: Choice of a simple float switch or a hygienic non-contact sensor.
- 1-Channel Relay Module (5V): Actuates the DC pump.
- Mini DC Water Pump: Pushes the liquid through the tubing.
- LEDs: For visual feedback.
- Container & Hose: For storing and guiding the alcohol.

Flow Chart

